Urban Traffic Facilities Design Project
(KFT07090)

Project Report on Redesign of a local access road (Wildenfelser Str.) within the city of Zwickau

SECTION: Intersection Muldestraße to Town Sign ("Am Wasserturm")

SUBMITTED BY: Group 1

Kevin Vekariya (MatrikelNr:61950)

Muhammad Hamza Khan (MatrikelNr:61692)

Aliyu Ayuba (MatrikelNr:61702)

INSTITUTION: WHZ Westsächsische Hochschule Zwickau Hochschule für Mobilität Study Programme Road Traffic Engineering / Mobilität und Verkehr Summer Term 2026

WHZ Westsächsische Hochschule Zwickau
Hochschule für Mobilität

CONTENTS

1. Introduction

The geometric design and spatial reorganization of urban road spaces are fundamental elements of traffic engineering. They are essential for balancing the competing demands of traffic capacity, municipal functionality, and environmental livability, while fundamentally maximizing traffic safety for all road users.

This project report details the engineering analysis, conceptual redesign, and geometric layout of a critical urban section of Wildenfelser Straße (S 283) located within the city of Zwickau. The planning, evaluation, and engineering drafting tasks were executed during the Summer Term 2026 within the framework of the academic module *KFT09070 Urban Facilities Design Project* at the Westsächsische Hochschule Zwickau.

The core objective of this assignment is to develop a highly functional, concept-driven reorganization of the existing space along Wildenfelser Straße. The redesign spans start from the heavily trafficked intersection at Muldestraße to the Zwickau town limits near the "Am Wasserturm" landmark.

The technical engineering solutions, cross-sectional distributions, and structural intersections developed during this project have been drafted and finalized utilizing AutoCAD. All design steps, structural dimensions, and control configurations strictly adhere to valid German guidelines and technical regulations. These primarily include the *Richtlinien für die Anlage von Stadtstraßen* (RASt) and the *Richtlinien für Lichtsignalanlagen* (RiLSA) and after town limit of Zwickau, the *Richtlinien für integrierte Netzgestaltung* (RIN).

2. Project and Site Description

- **Road Category Group:** Built-up main Urban Road / Local Access Arterial Road
- **Road Class / Designation:** State Road (S 283)
- **Planning Limits:** Intersection Muldestraße (Inbound) to City Boundary / Town Sign "Am Wasserturm" (Outbound)
- **Connecting Function:** Links the dense residential areas of the Oberhohndorf district to the core urban network of Zwickau and serves as a vital arterial conduit to the regional highway infrastructure via the A 72 (Zwickau Ost) interchange.

Traffic Parameters and Design Justification:

- **Forecasted Traffic Volume (2030):** 14,000 vehicles per 24 hours.
- **Heavy Traffic Share (HDV):** 7.8%.

The targeted planning section features a highly complex operational profile. Due to the geographical convergence of localized municipal origin/destination traffic with heavy volumes of inter-regional through traffic, the road corridor experiences significant operational pressure.

The existing infrastructure space is disproportionately optimized toward the requirements of private motorized vehicles. This leaves severe deficits in safety, quality of service, and accessibility for active, non-motorized traffic modes—specifically pedestrians and cyclists. Consequently, an extensive spatial reorganization is technically justified to maintain an adequate Level of Service (LOS) for motorized vehicles while comprehensively upgrading the surrounding pedestrian, cycling, and public transit facilities.

2.1. Analysis of Current Situation and Functional Conflicts

A meticulous engineering evaluation and spatial conflict analysis of Wildenfelser Straße reveals severe infrastructural deficiencies that heavily compromise traffic safety:

- **Cycling Infrastructure Deficits:**
  There are currently no dedicated cycling facilities within the urban planning boundaries. Cyclists are forced to navigate in mixed traffic directly alongside heavy vehicles under high-volume, high-speed conditions. This is further compounded by a demanding longitudinal gradient, presenting a critical, high-risk safety hazard. The existing segregated bicycle path terminates abruptly at the municipal boundary line near the Birkenweg / Lößnitzer Straße intersection in Reinsdorf. This leaves a dangerous infrastructural gap that must be bridged.
- **Pedestrian Accessibility Constraints:**
  The corridor exhibits high transverse pedestrian crossing demands generated by adjacent residential zones. However, there is a total absence of structural crossing aids, pedestrian islands, or marked crosswalks. This heavily limits barrier-free mobility and safe movement across the street.
- **Junction and Flow Inefficiencies:**
  Existing intersections lack dedicated left-turn lanes. Left-turning vehicles frequently block the shared lanes, causing severe traffic queues, rear-end collision risks, localized delays, and elevated vehicle emissions.
- **Environmental and Speed Disadvantages:**
  Motorized vehicle operating speeds remain inappropriately high, while elevated traffic noise continuously degrades the ambient quality of the adjacent urban environment.

2.2. Cross-Section Design and Variant Development

To resolve the severe structural and functional conflicts identified along Wildenfelser Straße (S 283), two distinct cross-sectional design variants have been modeled within AutoCAD. Both options are engineered to process the projected traffic load of 14,000 vehicles per 24 hours while systematically redistributing the available street width according to the *Richtlinien für die Anlage von Stadtstraßen* (RASt).

Technical Baseline Parameters:

- **Design Speed:** 50(km/h) (Urban Core) / 100(km/h) (Rural Transition Zone).
- **Design Vehicle:** Two-axle heavy commercial truck / standard public transit bus (critical for the 7.8% heavy traffic share).
- **Total Available Width (Property Line to Property Line):** Variable bottleneck constraint of approx. 15.50 to 17 meters.

Variant 1:

This layout focuses on narrowing the standard vehicle lanes to the minimum dimensions permitted by RASt for local access arterial roads. Reclaiming this lateral space allows for the integration of continuous, standard-compliant cycling infrastructure and widened pedestrian sidewalks on both sides of the roadway.

Cross-Section Details

This alternative focuses on the enhanced safety and protection of vulnerable road users by introducing dedicated safety buffers between active transport paths and motorized lanes.

- **Total Width:** 16.10 meters
- **Sidewalk (Both side):** 2.3 meters
- **Bicycle Lane (Both side):** 2.0 meters
- **Safety Buffer / Separation Strip (Both side):** 0.5 meters
- **Motorized Carriageway Lane (Both side)**: 3.25 meters

Variant 2:

This layout establishes a multi-modal spatial allocation featuring asymmetric elements. It incorporates dedicated, protected cycling tracks alongside specific center-lane zones designated for structural traffic islands and refuge areas. Variant 2 is the combination in which a single-side two-way cycle track (Zw-R) is carried on the south side across the route.

Cross-Section Details

- **Total Cross-Section Width:** 15.50 meters
- **Sidewalk / Footpath (North Side):** 2.50 meters
- **Motorized Travel Lane (Both Side):** 3.25 meters
- **Safety strip (South Side):** 0.75 meters
- **Two-way cycle track (South Side):** 3.25 meters
- **Boundry strip (South Side):** 0.30 meters
- **Sidewalk / Footpath (South Side):** 2.2 meters

Preferred Alternative Selection:

"Variant 2 has been chosen as the preferred alternative due to its superior spatial consistency throughout the entire Wildenfelser Straße corridor layout. By providing symmetrical sidewalks and continuous 3.25 m one side two-way cycle track, it establishes a reliable layout that fits predictably within the constrained street limits and helps to increase the speed limits of carriageway and cycle track. This approach simplifies construction, maximizes pedestrian safety, and ensures that dedicated cycling infrastructure remains fully continuous alongside the 6.5m vehicle carriageway."

2.3. Cycling and Pedestrian Infrastructure Integration

The integration of non-motorized facilities prioritizes safety, continuity, and barrier-free design across the entire planning section.

- **Cycling Infrastructure:** Within the urban limits, cycling facilities isolate cyclists from heavy vehicle currents, minimizing the risks associated with mixed traffic on steep gradients. At the rural boundary, a smooth transition connects the new lanes directly to the Reinsdorf municipal bicycle path.
- **Pedestrian Crossing Facilities:** To address high transverse crossing demands, tactile, barrier-free pedestrian refuge islands are structurally integrated into the center lane. These islands shorten crossing distances for pedestrians while acting as physical entry gateways that naturally lower inbound vehicle speeds.
- **Public Transport Accessibility:** Existing bus stops were evaluated for functionality. Boarding platforms have been redesigned with raised kerbs and tactile paving to ensure fully accessible, barrier-free boarding.

Both design variants maintain an effective roadway cross-section capable of safely accommodating the 14,000 Veh/24h traffic load. They also integrate a rural transition zone to systematically close the infrastructure gap between the Zwickau city and the existing Reinsdorf municipal bicycle path.

2.4. Junction Optimization and Left-Turn Lanes

To maintain an adequate Level of Service (LOS) and minimize the risk of rear-end collisions, specialized junction layouts were developed.

- **Structural Left-Turn Pockets:** Dedicated turn lanes are integrated into major intersection approaches. Removing decelerating or stationary left-turning vehicles from the main through lanes stabilizes traffic flow, reduces queuing, and lowers vehicle emissions caused by stop-and-go driving.
- **Signal Control Program:** For the signalized intersection within the project limits, a demand-oriented signal control program was designed.

- **Conflict Matrix & Phase Stages:** A technical conflict matrix was constructed to map all intersecting vehicle and pedestrian movements. This matrix established the exact intergreen safety times required to arrange compliant phase stages according to German *Richtlinien für Lichtsignalanlagen* (RiLSA) standards, maximizing green time efficiency for the 14,000 Veh per 24h volume.

Types of Junctions:

In both Variants, type of junction is selected according to the requirements of pedestrians and cyclists, width at the intersection point, and number of accidents.

1. Main Signalized Intersections:

Motorized Approaches: The regular **3.25 m** lanes **expand or split** into dedicated left-turn, through, and right-turn lanes where spatial constraints permit, maintaining a minimum operating lane width of **3 m** within the immediate intersection zone to handle heavy commercial vehicles and transit buses safely

2. Right-before-Left Junctions (*Rechts-vor-Links*):

In Germany, this is the default statutory priority rule at intersections according to **§ 8 Abs. 1 StVO** (German Traffic Regulations). It is heavily utilized in urban planning to manage traffic calmly without the need for signals or signs.

Application Criteria: It applies strictly where there are no traffic lights, traffic police, or priority-regulating traffic signs (such as Sign 306 🟩 "Priority Road" or Sign 205 🔻 "Yield").

3. Queuing Zones (*Aufstellbereiche / Warteschlangenbereiche*):
A queuing zone is a dedicated segment of an approach lane at an intersection designed to temporarily store vehicles waiting for a green light or a gap in traffic without blocking upstream movements.

2.5. Specialized Crossing Layouts & Safety Infrastructure

To ensure absolute continuity for active transport and maintain a high safety index within the structurally consistent 15.5-meter layout, junctions are optimized using **four standardized crossing types** aligned with German **RASt 06** and **RiLSA** guidelines:

Pedestrian Crossings (*Zebrastreifen* / Standard Crosswalks):

- **Variant 2:**
  On Main Road: 2 m
  On Side Road: 3 m

Cycle Crossings (*Radfahrerfurten*):

- **Variant 2:**
  On Side Road: 3 m (Continuous between Cycle tracks of 2 sections with Priority.)

Shared Crossings (*Gemeinsame Furten*):

- **Variant 1&2:**
  On Main & Side Road: 4.5 m (Combines the pedestrian crosswalk and the cycle crossing into a single parallel facility without a barrier.)

2.6. AutoCAD Drafting Layout and Technical Representation

The conceptual variants and detailed intersection plans were meticulously drafted within an AutoCAD environment. The technical drawing files compile all horizontal parameters, structural dimensions, lane allocations, and site boundaries into a cohesive layout.

Particular engineering attention was dedicated to the line weight definitions, layer hierarchy, and plotting configurations. Symbols for traffic control, crosswalk markings, and structural lane separation lines were scaled precisely to ensure that PDF exports remain legible, sharp, and standard-compliant across standard drawing sheet sizes. The complete AutoCAD documentation includes integrated cross-sectional profiles detailing the multi-modal space reallocation alongside the signal layout plans.

Standard Legend Information (From Drawing Layout):

To ensure AutoCAD drawing documentation matches perfectly, here is how the elements are categorized in drawing layout standard:

Color Coding System

- Driving Lanes: Plotted in light grey
- Cycle Lanes: Plotted in red
- Footpaths / Sidewalks: Plotted in brown
- Green Areas: Plotted in green

Key Intersection Features Modeled

- Signalized Intersection Interface: Standard layout managed under German signal design guideline (RiLSA).
- Crossings: Includes explicit markings for Pedestrian Crossings (Zebrastreifen), Cycle Crossings, and Shared Crossings.

2.7. Comparative Evaluation of Design Variants

To systematically determine the most optimal engineering solution for the reorganization of the Wildenfelser Straße (S 283) corridor, a comparative evaluation between Variant 1 and Variant 2 was performed. Both alternatives were strictly cross-examined against core key performance indicators (KPIs) including geometric network consistency, active transport safety, pedestrian space quality, and structural feasibility within the available property line limits.

Variant 2 is selected as the preferred design alternative for the following technical reasons:

1. **Geometric Uniformity:** Variant 2 maintains a highly predictable, mathematically consistent cross-section layout of exactly 15.5 meters along the entire planning scope. This consistency simplifies pavement construction and ensures clear sightlines for approaching drivers.
2. **Pedestrian Prioritization:** Expanding both flanking sidewalks to a spacious 2.5& 2.2 meters maximizes the pedestrian Level of Service (LOS), creating accessible walking infrastructure that handles high local pedestrian demands seamlessly.
3. **Continuous Cycling Corridor:** Variant 2 preserves complete cycling continuity. The red-marked cycle tracks run uninterrupted through the corridors and connect directly to the 3-meters wide designated cycle crossings (*Radfahrerfurten*) at intersections.
4. **Frictionless Vehicle Capacity:** By maintaining standard 3.25-meter travel lanes (6.5 meters total motorized carriageway), Variant 2 safely processes the heavy traffic demands of the state road without causing queuing spillbacks into upstream junctions.

3. Conclusion

The conceptual design and engineering analysis demonstrate that the current multi-modal deficits of Wildenfelser Straße (S 283) can be effectively resolved through a systematic reorganization of the cross-section space. Both variants developed in AutoCAD comply with the rigorous technical demands of the RASt and relevant German design standards. They ensure that the forecasted traffic volume of 14,000 vehicles per 24 hours can be processed with an acceptable Level of Service.

No critical structural compromises were needed to achieve standard compliance. Variant 2 emerges as the preferred layout for the project area because of its consistency throughout the route and safety for everybody. By incorporating dedicated cycling tracks, barrier-free pedestrian crossing islands, and localized left-turn lanes, it directly resolves the core safety conflicts of the corridor.

This design successfully bridges the infrastructural gap to the Reinsdorf municipal bicycle path while providing a functional, safe, and balanced urban road environment for all modes of transportation.

4. References

- Forschungsgesellschaft für Straßen- und Verkehrswesen (FGSV). (2006). *Richtlinien für die Anlage von Stadtstraßen (RASt 06)*.
- Forschungsgesellschaft für Straßen- und Verkehrswesen (FGSV). (2010). *Empfehlungen für Radverkehrsanlagen (ERA)*.
- Forschungsgesellschaft für Straßen- und Verkehrswesen (FGSV). (2012). *Richtlinien für integrierte Netzgestaltung (RIN)*.
- Forschungsgesellschaft für Straßen- und Verkehrswesen (FGSV). (2015). *Richtlinien für Lichtsignalanlagen (RiLSA)*.
- Ministerium für Verkehr Baden-Württemberg (VM BW). (2020).
- Autodesk, Inc. (2026). *AutoCAD Mechanical 2026 / Design Environment software*.
- *Course materials: KFT09070 Urban Traffic Facilities Design Project,* Road Traffic Engineering, Westsächsische Hochschule Zwickau.